0100. Same Tree

Easy · Trees

leetcode.com/problems/same-tree

Problem

- Return true if two binary trees have identical structure and values.

Intuition

- Two trees are equal if their roots match and their left subtrees are equal and their right subtrees are equal.

Brute Force

- Serialize both (with null markers) and compare strings: O(n) but extra memory.

Optimal Approach

- Both null → true. One null → false.
- Values differ → false.
- Return same(p.left, q.left) && same(p.right, q.right).

Worked Example

```
p = [1,2], q = [1,null,2].

roots 1 == 1
left: p.left=2, q.left=null -> false
```

Java Solution

```java
public class SameTree {
    public boolean isSameTree(TreeNode p, TreeNode q) {
        if (p == null || q == null) return p == q;
        return p.val == q.val && isSameTree(p.left, q.left) && isSameTree(p.right, q.right);
    }
}
```

Complexity

- Time: O(min(n, m)). Space: O(h).

Pitfalls

- Handle the null cases before reading .val.